

BHAKTI MARAKANA

Senior Software Engineer

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PROFESSIONAL SUMMARY

Innovative Sr. Software Engineer with over 4 years of professional experience in the full software development lifecycle. Expert in designing, developing, and managing complex web applications using Python and PHP frameworks including Django, Flask, Magento, Laravel, and Zend. Proven ability to architect high-performing, secure, and scalable backend systems in e-commerce, travel, and fintech domains. Expert in optimizing database systems and developing robust API architectures for mobile and third-party integrations. Available for immediate joining for Python-centric roles.

TECHNICAL SKILLS

Backend Development: Python, Django, Flask, Machine Learning(Basic), PHP, Magento 2, Laravel, Zend, Node.js.

APIs & Integrations: RESTful API Development, Third-party API Integration, OAuth2 Authentication.

Database Systems: MySQL, PostgreSQL, Advanced Query Optimization.

Frontend Tech: JavaScript (ES6+), jQuery, Ajax, HTML5, CSS3.

Tools & DevOps: Git, Github, Jira, Postman.

Core Expertise: Task Scheduling (CRON), Dynamic Analytics Dashboards, Performance Optimization, Scalable Systems.

PROFESSIONAL EXPERIENCE

Sr. Software Engineer

08/2020 – 10/2024

Cybercom Creation, Ahmedabad, Gujarat

- Architected and maintained high-traffic backend systems using Magento, Laravel, and Zend, focusing on reusability and clean architecture.
- Developed robust RESTful APIs to serve as the backbone for mobile applications and complex third-party integrations across multiple domains.
- Strategically optimized database performance to ensure long-term maintainability and rapid scalability for growing client platforms.
- Collaborated within Agile/Scrum environments, leading feature implementations from conceptual design to high-quality production deployment.
- Improved system security and reliability by implementing advanced OAuth2 authentication and real-time notification queues via Firebase.
- Monitored and enhanced server-side performance through efficient backend logic and optimized SQL database query structures.

TECHNICAL PROJECTS

Disease Prediction using AI (Flask)

- Engineered a comprehensive full-stack AI application designed to analyze complex patient symptom profiles and predict health conditions with high statistical accuracy.
- Implemented and fine-tuned a Decision Tree classification algorithm using Scikit-Learn, utilizing optimized train/test dataset splits to validate performance.
- Designed and executed rigorous data preprocessing, cleaning, and feature engineering to transform raw medical symptoms into structured data.
- Built a responsive web interface using the Flask framework, enabling users to input varied symptoms and receive immediate, data-driven diagnostic probabilities.
- Applied cross-validation techniques and model evaluation metrics (Accuracy, Precision, Recall) to ensure reliability across diverse patient datasets.

Diabetes Risk Prediction(Flask)

- Full-stack ML application for predicting diabetes risk from patient health metrics, built on the Pima Indians Diabetes Database and deployed as a live web service.
- Performed EDA and data cleaning on 768 patient records, identifying biologically implausible zero-values in Glucose, BMI, and other clinical features and correcting them via median imputation
- Trained and benchmarked 4 classification models (Logistic Regression, k-NN, Random Forest, XGBoost), selecting XGBoost based on F1-score and false-negative minimization for clinical risk-sensitivity
- Tuned the model with GridSearchCV across a 5-parameter grid (243 combinations, 5-fold CV) to optimize F1-score
- Applied SHAP (SHapley Additive exPlanations) for model interpretability, generating feature importance rankings and dependence plots to explain individual predictions
- Packaged preprocessing (imputation, scaling) and the tuned classifier into a single scikit-learn Pipeline, serialized with joblib for production use
- Built a Flask REST API (/predict endpoint) serving real-time predictions with class labels and confidence scores, with CORS support for cross-origin requests
- Designed a multi-step wizard UI with a dark navy glassmorphic aesthetic and an SVG risk gauge for visualizing prediction confidence
- Deployed on Render.com with Gunicorn as the WSGI serverTech

NHY Travel - 2ezasia (Django)

- Engineered a high-performance, scalable travel backend using the Django framework, specifically designed to handle high concurrent traffic for global travel bookings.
- Architected and deployed custom modules for Travel Management, Category Sorting, Travel Itineraries, Transport Logistics, and Event CMS integration.
- Designed and implemented a sophisticated dynamic dashboard providing real-time data visualization for business analytics, travel statistics, and administrative oversight.
- Built a suite of robust RESTful APIs for mobile application synchronization and automated CRON systems to handle recurring travel status updates and system tasks.

Point of Sale - Gosolution(Laravel)

- Architected core POS functional modules including real-time inventory tracking, multi-tier product management, and complex digital voucher and reward point systems.
- Engineered a dynamic CRON management system to automate mission-critical business processes, including financial reconciliation and automated reporting.
- Integrated Firebase for real-time data synchronization and push notifications, ensuring seamless updates and consistency across all connected POS clients.
- Designed optimized commission logic, appointment scheduling, and voucher modules tailored to client-specific workflows and financial regulations.

1Stopbedrooms (Magento + Python)

- Led the development of specialized Magento extensions, including custom shipping method logic, AJAX-integrated cart updates, and highly customized checkout workflows.
- Developed a suite of Python scripts to bridge Magento backend operations with automated processes, focusing on log analysis and automated data integrity checks.
- Significantly boosted system efficiency by implementing advanced server-side caching, database query optimization, and lazy loading techniques to improve SEO.
- Designed and deployed multi-layered reporting dashboards featuring complex filters and search functionalities to provide clients with granular business analytics and sales insights.

CERTIFICATIONS

- Introduction to Machine Learning Concepts – Microsoft
- Neural Networks and Deep Learning - DeepLearning.AI
- Improving Deep Neural Networks: Hyperparameter Tuning, Regularization and Optimization - DeepLearning.AI
- Deloitte Australia - Data Analytics Job Simulation - Forage
- Build & Deploy Apps with Google AI Studio & Multilingual AI Speech App with Vibe Coding – GUVI

EDUCATION

Bachelor of Engineering in Computer Engineering

Government Engineering College, Gandhinagar

- CGPA: 8.62

08/2016 – 06/2020